Interventional causal representation learning

STATS 212: Graphical Models

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Motivation

Deep learning models (e.g., GPT-3, CLIP) have achieved remarkable success because of their ability to learn features (or representations) from data. However, it is often difficult for these models to perform new tasks and adapt to distribution shifts[1].

[1] Bommasani et al., *On the opportunities and risks of foundation models*, 2022

Motivation

A line of research known as *causal representation learning* (which involves augmenting these models with causal understanding) has emerged to address this shortcoming. In this presentation, we discuss a recent paper by Ahuja et al.[2], which characterizes the impact of interventional data on identifying latent factors.

[2] Ahuja et al., "Interventional causal representation learning", 2023

A brief overview

- Many works rely on observational data and make assumptions on the dependency structure of the latents to achieve some degree of identification.
- Key questions: Can interventional data help identify latent factors in causal representation learning? How can it help?
- An important observation is that interventional data often carries geometric signatures of the latent factors' support. The authors relax the stricter distributional assumptions and leverage that geometric information to show that identification can be achieved.

Setup & Notation

- Causal representation learning is tasked with identifying latents (e.g., coordinates) from high-dimensional observations (e.g., images).
- We assume access to both observational and interventional data.
  - Observations: $x \in \mathbb{R}^n$
  - Latents: $z \in \mathbb{R}^d$
- Observational data generating process:

$$z \sim \mathbb{P}_z, \quad x \leftarrow g(z)$$

- Latent z is sampled from distribution $\mathbb{P}_z$ and observation x is obtained by applying an injective "decoder" function $g : \mathbb{R}^d \to \mathbb{R}^n$ to z.

Setup & notation

- Interventional data generating process:

$$z \sim \mathbb{P}_z^{(i)}, \quad x \leftarrow g(z)$$

  - Denote $\mathcal{Z}$ and $\mathcal{Z}^{(i)}$ as the support of $\mathbb{P}_z$ and $\mathbb{P}_z^{(i)}$, respectively.
  - The support of x is $\mathcal{X} = g(\mathcal{Z})$ for observational data and $\mathcal{X}^{(i)} = g(\mathcal{Z}^{(i)})$ for interventional data.

Setup & notation

- Main goal of causal representation learning is to learn a so-called "encoder" function that takes in x and outputs its underlying latent z.
- In practice, this encoder is learned by solving a reconstruction identity,

$$h \circ f(x) = x \qquad \forall x \in \mathcal{X} \cup \mathcal{X}^{(i)}, \tag{1}$$

  where $f : \mathbb{R}^n \to \mathbb{R}^d$ and $h : \mathbb{R}^d \to \mathbb{R}^n$ are a pair of encoder and decoder, which need to jointly satisfy the reconstruction identity.
  - The pair (f, h) together is referred to as the autoencoder.
- Once we learn the encoder f, the resulting representation is $\hat{z} := f(x)$, which holds the encoder's estimate of the latents.

Assumptions

We have the following technical requirements, assumed to hold throughout.

Assumption 1

The interior of the support of z, $\mathcal{Z} \cup \mathcal{Z}^{(i)}$, is a non-empty subset of $\mathbb{R}^d$.

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Assumptions

Assumption 2

The (true) decoder g is a polynomial of finite degree p whose corresponding coefficient matrix G has full column rank.

Constraint

Constraint 1

The learned decoder h is a polynomial of degree p and it is determined by its corresponding coefficient matrix H as follows,

$$h(z) = H[1, z, z\bar{\otimes}z, \cdots, \underbrace{z\bar{\otimes}\cdots\bar{\otimes}z}_{p \text{ times}}]^\top \qquad \forall z \in \mathbb{R}^d,$$

where $\bar{\otimes}$ represents the Kronecker product with all distinct entries. The interior of the image of the encoder $f(\mathcal{X} \cup \mathcal{X}^{(i)})$ is a non-empty subset of $\mathbb{R}^d$.

Main result 1

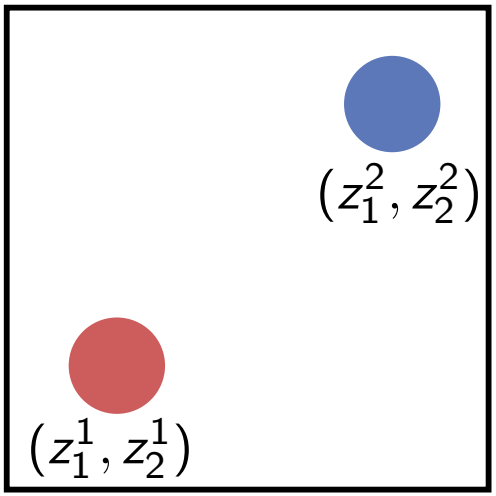
Theorem 1 (Affine identification of latents)

Suppose the observational data and interventional data is generated from $z \sim \mathbb{P}_Z$ *and* $z \sim \mathbb{P}_Z^{(i)}$*, respectively. The autoencoder that solves the reconstruction identity* $h \circ f(x) = x$ *under Constraint 1 achieves affine identification, i.e.,* $\forall z \in \mathcal{Z} \cup \mathcal{Z}^{(i)}$, $\hat{z} = Az + c$*, where* $\hat{z}$ *is the encoder* f*'s output,* z *is the true latent,* $A \in \mathbb{R}^{d \times d}$ *is invertible and* $c \in \mathbb{R}^d$*.*

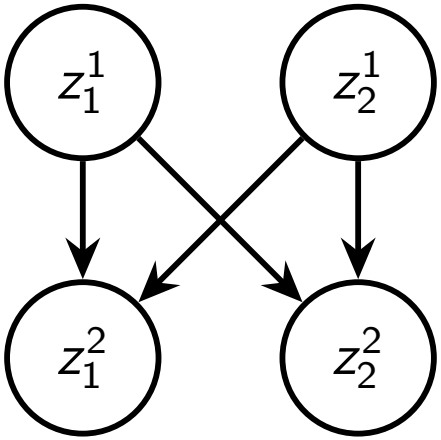
Remarks

- Theorem 1 reduces ambiguities in identifying z from arbitrary invertible transformations to only invertible affine transformations.
- Moreover, strong assumptions about the structure of the latents is not required.

Motivating example

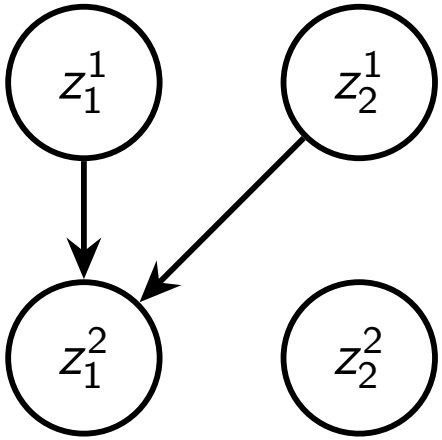


(a)



Observational Data

(b)



Interventional Data

(c)

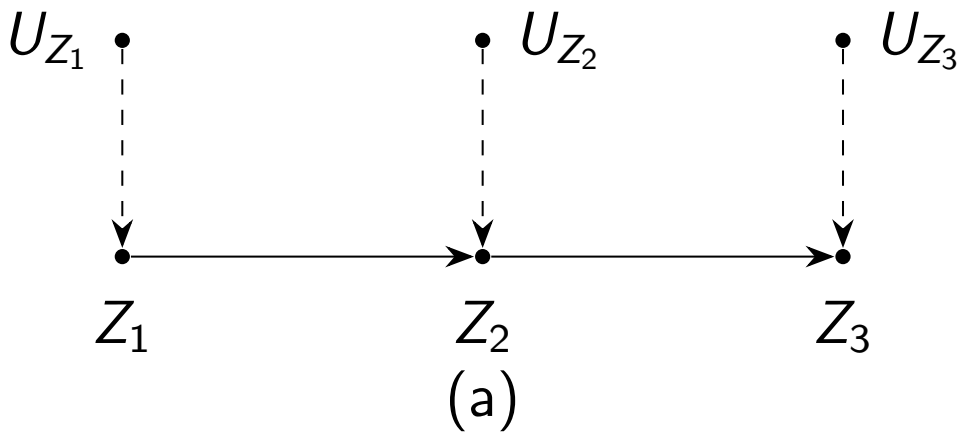
More preliminary setup

- Now consider the addition of the *do* intervention to the causal representation learning problem
- Formally, a *do* intervention on one latent dimension fixes it to some constant value.
  - As such, the distribution of the children of the intervened component is affected by the intervention, while the distribution of remaining latents remains unaltered.
- Based on this discussion, we have the following notation:

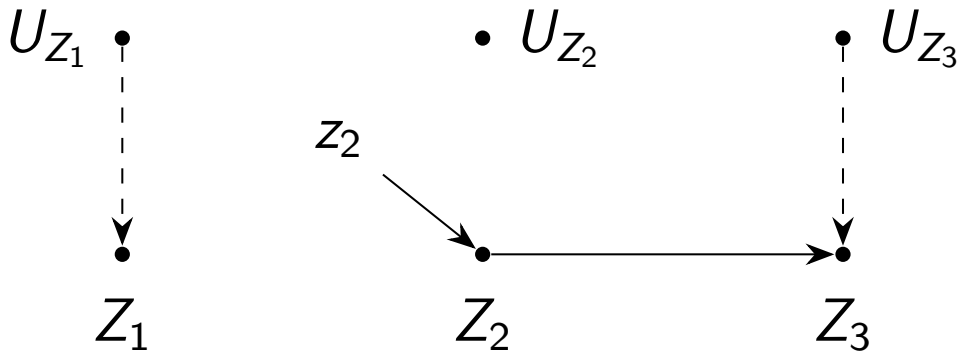
$$z_i = z^*; \qquad z_{-i} \sim \mathbb{P}^{(i)}_{Z_{-i}},$$

where z_i takes a fixed value z^* based on the *do* intervention and the remaining variables z_{-i} are sampled from the distribution $\mathbb{P}^{(i)}_{Z_{-i}}$.

More preliminary setup



(a)



(b)

FIG 1. (a) Toy example of a latent variables diagram. (b) The diagram associated with the modified model under the intervention $do(Z_2 = z_2)$.

Constraint

Constraint 2

The encoder's k-th component $f_k(x)$ denoted as $\hat{z}_k$ is required to take some fixed value $z^\dagger$ for all $x \in \mathcal{X}^{(i)}$, i.e., $f_k(x) = z^\dagger, \forall x \in \mathcal{X}^{(i)}$.

This constraint is not too restrictive, since we do not need to know which component is intervened and the value it takes.

Main result 2

Theorem 2 (Identification under intervention)

Suppose the observational data and interventional data are generated from $z \sim \mathbb{P}_Z$ *and* $z \sim \mathbb{P}_Z^{(i)}$*, respectively under Assumptions 1 and 2, where* $\mathbb{P}_Z^{(i)}$ *follows* $z_i = z^*; z_{-i} \sim \mathbb{P}_{Z_{-i}}^{(i)}$*. The autoencoder that solves the reconstruction identity* $h \circ f(x) = x$ *under Constraints 1 and 2 identifies the intervened latent* z_i *up to shift and scaling, i.e.,* $\hat{z}_k = ez_i + b$*,* $e, b \in \mathbb{R}$*.*

Remarks

- Theorem 2 extends to settings when multiple interventional distributions are available, with each corresponding to a hard *do* intervention on a distinct latent variable.
- It does not rely on any distributional assumptions on z nor on the structure of the graphical model.
- This result can be generalized to non-polynomial g: the intervened latent can be approximately identified up to an invertible transform (but one *do* intervention per latent is no longer enough).

Perfect interventions

- Under a "perfect" intervention, a latent that is under intervention are fully disconnected from its parents
  - A *do* intervention is an example of a perfect intervention
- Perfect interventions make the intervened latent independent of all of its non-descendants.

Imperfect interventions

- On the other hand, imperfect interventions do not completely remove the influence of the parents on the intervened latent, but can instead reduce it.
- Thus, imperfect interventions are not guaranteed to lead to independence between random variables. More explicitly, the intervened variable is not guaranteed to be independent of its non-descendants.
- Key insight: these imperfect interventions do not necessarily need to induce independence between the latents; rather, they merely need independence of supports (IOS) between the latents z_i and z_j, $i \neq j$.
- This naturally led the authors to specifically consider imperfect interventions that lead to support independence, which ties in nicely with Assumption 3.

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- This naturally led the authors to specifically consider imperfect interventions that lead to support independence, which ties in nicely with Assumption 3.

Assumption

Assumption 3

Consider z sampled from the interventional distribution $\mathbb{P}_Z^{(i)}$, where $\mathbb{P}_Z^{(i)}$ follows $z_i = z^*; z_{-i} \sim \mathbb{P}_{Z_{-i}}^{(i)}$. There exists $\mathcal{S} \subseteq [d]$ such that the support of z_i is independent of z_j for all $j \in \mathcal{S}$. For all $j \in \mathcal{S}$

$$\mathcal{Z}_{i,j}^{(i)} = \mathcal{Z}_i^{(i)} \times \mathcal{Z}_j^{(i)}.$$

For all $j \in [d], -\infty < \alpha_{\text{inf}}^j \leq \alpha_{\text{sup}}^j < \infty \exists \zeta > 0$ such that $(\alpha_{\text{sup}}^j - \zeta, \alpha_{\text{sup}}^j) \cup (\alpha_{\text{inf}}^j, \alpha_{\text{inf}}^j + \zeta) \subseteq \mathcal{Z}_j^{(i)}, \forall j \in [d]$.

Constraint

Constraint 3

For each $m \in \mathcal{S}'$ (where $\mathcal{S}'$ is a set) $(\hat{z}_k, \hat{z}_m)$ satisfies support independence on interventional data, i.e.,

$$\hat{\mathcal{Z}}_{k,m}^{(i)} = \hat{\mathcal{Z}}_k^{(i)} \times \hat{\mathcal{Z}}_m^{(i)}, \quad \forall m \in \mathcal{S}'.$$

Main result 3

Theorem 3 (Block affine identification)

Suppose the observational data and interventional data are generated from $z \sim \mathbb{P}_Z$ *under Assumptions 1, 2, and 3. The autoencoder that solves Eq.* (1) *under Constraint 1 and Constraint 3 achieves block affine identification. More specifically,* $\forall z \in \mathcal{Z} \cup \mathcal{Z}^{(i)}$

$$\hat{z}_k = a_k^\top z + c_k, \hat{z}_m = a_m^\top z + c_m, \forall m \in \mathcal{S}',$$

where a_k *contains at most* $d - |\mathcal{S}'|$ *non-zero elements and each component of* a_m *is zero whenever the corresponding component of* a_k *is non-zero for all* $m \in \mathcal{S}'$.

Remarks

- Theorem 3 implies that $\hat{z}_k$ does not depend on all the latents.
- Observe that if the subset S' is large enough - specifically, $d-1$ dimensions, then Theorem 3 reduces to Theorem 2. Namely, $\hat{z}_k = cz_j + a$, for some $c, a \in \mathbb{R}$.

Special case

Consider the special case where the support of the latents in the observational data already satisfies the IOS condition without interventions. Following the previous sections, we have the following constraint where the learner leverages the geometric structure in the support to search for the autoencoder.

Constraint 4

Each pair $(\hat{z}_k, \hat{z}_m)$, where $k, m \in [d]$ and $k \neq m$ satisfies support independence on observational data.

Special case

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Constraint 4

Each pair $(\hat{z}_k, \hat{z}_m)$*, where* $k, m \in [d]$ *and* $k \neq m$ *satisfies support independence on observational data.*

Main result 4

Theorem 4 (Identification with independent support)

Suppose the observational data is generated from $z \sim \mathbb{P}_Z$ *under Assumptions 1 and 2. Further assume that the latents satisfy pairwise support independence. The autoencoder that solves Eq.* (1) *under Constraint 4 achieves permutation, shift, and scaling identification, i.e.,* $\forall z \in \mathcal{Z}$, $\hat{z} = \Lambda \Pi z + c$, *where* $\hat{z}$ *is the output of the encoder* f, z *is the true latent,* Π *is a permutation matrix, and* Λ *is an invertible diagonal matrix.*

Remarks

- Theorem 4 states that independence between the latents' support is sufficient to achieve identification.
- Moreover, even if the latent variables are dependent, the latent variables can be identified up to permutation, shift and scaling, as long as they are bounded (hence non-Gaussian) and satisfy pairwise support independence.

Non-polynomial mappings

- One of the main restrictions of the results stated thus far is that the mapping is a polynomial.
- These results can be generalized to non-polynomial g: if the number of *do* interventional datasets is sufficiently large and the support of the latents satisfy certain regularity conditions, then the autoencoder approximately identifies the latent up to an invertible transformation.

Training procedure

Algorithm 1 Summary of two-step approach (IOS objective)

Step 1: Train autoencoder (f, h)

Sample data: $X \sim \mathcal{X} \cup \mathcal{X}^I$ where $\mathcal{X}^I = \cup_{i=1}^d \mathcal{X}^{(i)}$

Minimize reconstruction loss: $f^\dagger, h^\dagger = \arg\min_{f,h} \mathbb{E}\left[\|h \circ f(X) - X\|^2\right]$

Step 2: Learn transformation Γ with IOS objective

Sample data: $\hat{Z} \sim f^\dagger(\mathcal{X})$ where $f^\dagger$ is the encoder learnt in Step 1

Minimize reconstruction + Hausdorff loss:

$$\min_{\Gamma} \mathbb{E}\left[\|\Gamma' \circ \Gamma(\hat{Z}) - \hat{Z}\|^2\right] + \lambda \times \sum_{k \neq m} \mathrm{HD}(\hat{\mathcal{Z}}_{k,m}(\Gamma), \hat{\mathcal{Z}}_k(\Gamma) \times \hat{\mathcal{Z}}_m(\Gamma))$$

Return transformed latents: $\Gamma(\hat{Z})$

Training procedure

Algorithm 2 Summary of two-step approach (interventional data)

Step 1: Train autoencoder (f, h)

Sample data: $X \sim \mathcal{X} \cup \mathcal{X}^I$ where $\mathcal{X}^I = \cup_{i=1}^d \mathcal{X}^{(i)}$

Minimize reconstruction loss: $f^\dagger, h^\dagger = \arg\min_{f,h} \mathbb{E}\left[\|h \circ f(X) - X\|^2\right]$

Step 2: Learn transformation $\Gamma = [\gamma_i]_{i=1:d}$ **using do-interventions**

for i in $\{1, ..., d\}$ **do**

Sample data: $\hat{Z} \sim f^\dagger(\mathcal{X})$ where $f^\dagger$ is the encoder learnt in Step 1

Fix intervention targets at random: $\hat{Y}^{(i)} \sim \mathcal{U}_{[0,1]}$

Minimize MSE loss: $\min_{\gamma_i} \mathbb{E}_{\hat{Z}}\left[\|\gamma_i(\hat{Z}) - \hat{Y}^{(i)}\|^2\right]$

end for

Return transformed latents: $\Gamma(\hat{Z})$

Model parameters

For all experiments, the authors follow the aforementioned two-stage training procedure. The parameters are as follows:

- Encoder: MLP with 3 hidden layers and 200 nodes per layer.
- Learn a polynomial decoder h (when g is a polynomial).
- To verify affine identification: R^2.
- To verify permutation, shift, and scaling identification: mean correlation coefficient (MCC).

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Experiment setup

Data is sampled according to the process in assumption 2 with the following details:

- Latent dimension: $d \in \{6, 10\}$.
- Degree of decoder polynomial g: $p \in \{2, 3\}$.
- Data dimension: $n = 200$.
- Decoder polynomial coefficient matrix G: $G_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $i = [n], j = [q]$.

For experiments with interventional data, the index of the variable to be intervened is sampled from Unif$(\{1, ..., d\})$ and the value is set to 2.0.

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Experiment setup

The latent distributions ($\mathbb{P}_Z$) used in the experiments are as follows:

- Uniform: $z_i \overset{\text{i.i.d.}}{\sim} \mathcal{U}_{[-5,5]}$.
- Uniform-correlated: Consider a pair of latent variables z_i, z_{i+1} and sample two confounder variables $c_1 \sim \text{Bern}(0.5), c_2 \sim \text{Bern}(0.9)$. We sample z_i, z_{i+1} as follows:

$$z_i \sim \begin{cases} \mathcal{U}_{[0,0.5]} & \text{if } c_1 = 1 \\ \mathcal{U}_{[-0.5,0]} & \text{if } c_1 = 0 \end{cases},$$

$$z_{i+1} \sim \begin{cases} \mathcal{U}_{[0,0.3]} & \text{if } c_1 \oplus c_2 = 1 \\ \mathcal{U}_{[-0.3,0]} & \text{if } c_1 \oplus c_2 = 0 \end{cases}.$$

Experiment setup

The latent distributions ($\mathbb{P}_Z$) used in the experiments are as follows:

- Gaussian mixture:

$$z_i \sim \begin{cases} \mathcal{N}(0,1) & \text{with prob. } 0.5 \\ \mathcal{N}(1,2) & \text{with prob. } 0.5 \end{cases}.$$

- SCM-S: z is sampled as a DAG with d nodes using the Erdős-Rényi (ER) scheme with linear causal mechanism and Gaussian noise[3]. The expected density is set to 0.5.
- SCM-D: z is sampled as a DAG with d nodes using the same ER scheme as before, but with expected density set to 1.

[3] Brouillard et al., *Differentiable causal discovery from interventional data*, 2020

Results: Observational data (polynomial)

$\mathbb{P}_Z$	d	p	R^2	MCC	MCC (IOS)
Uniform	6	2	1.00 ± 0.00	66.91 ± 2.45	99.31 ± 0.07
Uniform	6	3	1.00 ± 0.00	75.14 ± 3.93	99.39 ± 0.06
Uniform	10	2	1.00 ± 0.00	58.49 ± 2.26	90.73 ± 2.92
Uniform	10	3	0.99 ± 0.00	56.77 ± 0.60	94.62 ± 1.50
Uniform-C	6	2	1.00 ± 0.00	71.19 ± 2.29	96.81 ± 0.11
Uniform-C	6	3	1.00 ± 0.00	70.53 ± 1.1	96.29 ± 0.05
Uniform-C	10	2	1.00 ± 0.00	64.58 ± 1.81	85.31 ± 2.35
Uniform-C	10	3	1.00 ± 0.00	62.69 ± 0.92	87.20 ± 1.77
Gaussian-Mixture	6	2	1.0 ± 0.0	70.53 ± 1.25	67.43 ± 2.01
Gaussian-Mixture	6	3	0.99 ± 0.0	66.19 ± 1.38	67.94 ± 1.42
Gaussian-Mixture	10	2	1.0 ± 0.0	59.5 ± 2.22	58.3 ± 0.67
Gaussian-Mixture	10	3	0.99 ± 0.0	57.15 ± 0.43	59.08 ± 1.11
SCM-S	6	2	0.96 ± 0.02	71.77 ± 1.43	72.61 ± 1.48
SCM-S	6	3	0.87 ± 0.07	73.14 ± 3.44	70.56 ± 1.54
SCM-S	10	2	0.99 ± 0.0	64.35 ± 2.0	65.86 ± 1.32
SCM-S	10	3	0.90 ± 0.05	61.95 ± 0.98	58.77 ± 1.27
SCM-D	6	2	0.97 ± 0.01	75.25 ± 2.85	61.61 ± 4.36
SCM-D	6	3	0.81 ± 0.11	82.9 ± 3.11	65.19 ± 2.70
SCM-D	10	2	0.83 ± 0.10	67.49 ± 2.32	69.64 ± 3.09
SCM-D	10	3	0.72 ± 0.15	66.4 ± 1.86	60.1 ± 1.16

Results: Interventional data (polynomial)

$\mathbb{P}_Z$	d	p	R^2	MCC	MCC (IL)
Uniform	6	2	1.0 ± 0.0	69.11 ± 1.11	100.0 ± 0.0
Uniform	6	3	1.0 ± 0.0	73.42 ± 0.49	100.0 ± 0.0
Uniform	10	2	1.0 ± 0.0	59.96 ± 2.03	100.0 ± 0.0
Uniform	10	3	1.0 ± 0.0	65.94 ± 0.80	99.85 ± 0.03
Uniform-C	6	2	1.0 ± 0.0	71.2 ± 2.46	100.0 ± 0.0
Uniform-C	6	3	1.0 ± 0.0	70.21 ± 1.90	99.97 ± 0.01
Uniform-C	10	2	1.0 ± 0.0	61.02 ± 1.03	100.0 ± 0.0
Uniform-C	10	3	1.0 ± 0.0	61.16 ± 1.59	99.91 ± 0.01
Gaussian-Mixture	6	2	1.0 ± 0.0	67.72 ± 2.20	99.99 ± 0.01
Gaussian-Mixture	6	3	0.99 ± 0.0	70.21 ± 2.74	99.39 ± 0.05
Gaussian-Mixture	10	2	1.0 ± 0.0	60.77 ± 1.60	99.98 ± 0.01
Gaussian-Mixture	10	3	0.99 ± 0.0	57.81 ± 1.16	99.46 ± 0.05
SCM-S	6	2	0.99 ± 0.0	68.41 ± 0.90	99.53 ± 0.38
SCM-S	6	3	0.99 ± 0.0	74.12 ± 2.32	99.25 ± 0.34
SCM-S	10	2	1.0 ± 0.0	68.01 ± 2.36	99.92 ± 0.03
SCM-S	10	3	0.97 ± 0.01	66.81 ± 1.10	98.8 ± 0.13
SCM-D	6	2	0.99 ± 0.01	71.8 ± 3.77	99.64 ± 0.12
SCM-D	6	3	0.98 ± 0.01	79.48 ± 3.45	98.22 ± 1.07
SCM-D	10	2	0.98 ± 0.01	70.78 ± 1.89	95.3 ± 2.24
SCM-D	10	3	0.97 ± 0.0	70.08 ± 2.80	97.24 ± 0.88

Results: Observational data (NN)

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Uniform	6	2	0.98 ± 0.0	73.75 ± 2.85	99.05 ± 0.02
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Uniform	10	3	0.94 ± 0.0	59.27 ± 1.06	91.24 ± 4.99
Uniform-C	6	2	0.96 ± 0.02	68.46 ± 1.94	94.95 ± 1.83
Uniform-C	6	3	0.91 ± 0.0	68.09 ± 1.56	89.14 ± 2.38
Uniform-C	10	2	0.99 ± 0.0	62.63 ± 2.05	88.88 ± 3.28
Uniform-C	10	3	0.91 ± 0.01	59.91 ± 1.75	81.76 ± 3.67
Gaussian-Mixture	6	2	0.95 ± 0.0	72.59 ± 2.03	65.33 ± 1.11
Gaussian-Mixture	6	3	0.84 ± 0.01	66.25 ± 2.86	63.43 ± 1.27
Gaussian-Mixture	10	2	0.95 ± 0.0	57.12 ± 1.52	54.76 ± 1.26
Gaussian-Mixture	10	3	0.83 ± 0.01	55.41 ± 1.40	52.87 ± 0.86
SCM-S	6	2	0.59 ± 0.18	69.77 ± 3.87	66.04 ± 1.34
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SCM-S	10	2	0.78 ± 0.08	65.99 ± 1.14	63.52 ± 1.11
SCM-S	10	3	0.11 ± 0.43	66.96 ± 2.60	62.11 ± 1.36
SCM-D	6	2	0.63 ± 0.22	75.19 ± 2.62	61.89 ± 4.0
SCM-D	6	3	0.47 ± 0.25	77.83 ± 3.49	65.85 ± 1.58
SCM-D	10	2	0.46 ± 0.11	69.81 ± 1.43	65.35 ± 2.72
SCM-D	10	3	0.10 ± 0.29	65.89 ± 2.56	61.92 ± 1.95

Results: Interventional data (NN)

$\mathbb{P}_Z$	d	p	R^2	MCC	MCC (IL)
Uniform	6	2	0.98 $\pm$ 0.0	68.39 $\pm$ 1.21	99.09 $\pm$ 0.02
Uniform	6	3	0.91 $\pm$ 0.0	63.2 $\pm$ 1.33	91.67 $\pm$ 2.50
Uniform	10	2	0.99 $\pm$ 0.0	60.54 $\pm$ 1.81	99.59 $\pm$ 0.04
Uniform	10	3	0.95 $\pm$ 0.0	61.0 $\pm$ 1.48	93.73 $\pm$ 0.45
Uniform-C	6	2	0.94 $\pm$ 0.02	65.91 $\pm$ 0.53	96.43 $\pm$ 1.47
Uniform-C	6	3	0.91 $\pm$ 0.0	67.92 $\pm$ 3.48	94.8 $\pm$ 0.28
Uniform-C	10	2	0.99 $\pm$ 0.0	60.02 $\pm$ 1.83	99.42 $\pm$ 0.01
Uniform-C	10	3	0.92 $\pm$ 0.0	61.68 $\pm$ 1.20	93.83 $\pm$ 0.78
Gaussian-Mixture	6	2	0.94 $\pm$ 0.0	70.64 $\pm$ 3.83	96.87 $\pm$ 0.14
Gaussian-Mixture	6	3	0.83 $\pm$ 0.01	64.43 $\pm$ 2.36	84.53 $\pm$ 2.60
Gaussian-Mixture	10	2	0.94 $\pm$ 0.0	57.35 $\pm$ 1.62	97.06 $\pm$ 0.16
Gaussian-Mixture	10	3	0.83 $\pm$ 0.0	57.89 $\pm$ 2.06	80.14 $\pm$ 1.77
SCM-S	6	2	0.94 $\pm$ 0.02	74.68 $\pm$ 2.28	93.07 $\pm$ 2.16
SCM-S	6	3	0.89 $\pm$ 0.02	71.56 $\pm$ 3.18	88.66 $\pm$ 2.71
SCM-S	10	2	0.98 $\pm$ 0.0	66.08 $\pm$ 1.04	94.14 $\pm$ 0.39
SCM-S	10	3	0.88 $\pm$ 0.01	63.35 $\pm$ 1.44	76.62 $\pm$ 6.15
SCM-D	6	2	0.95 $\pm$ 0.02	76.99 $\pm$ 2.53	91.63 $\pm$ 1.90
SCM-D	6	3	0.88 $\pm$ 0.01	75.72 $\pm$ 1.69	88.19 $\pm$ 3.63
SCM-D	10	2	0.95 $\pm$ 0.01	68.71 $\pm$ 2.16	90.14 $\pm$ 4.35
SCM-D	10	3	0.86 $\pm$ 0.01	68.52 $\pm$ 2.11	81.82 $\pm$ 3.0

Results: Image data

#interv dist.	Uniform	SCM linear	SCM non-linear
1	34.2 ± 0.24	12.8 ± 0.28	19.7 ± 0.31
3	73.9 ± 0.38	73.2 ± 0.33	59.7 ± 0.28
5	73.6 ± 0.21	83.4 ± 0.21	62.8 ± 0.2
7	72.5 ± 0.34	84.2 ± 0.25	69.3 ± 0.34
9	73.1 ± 0.47	86.2 ± 0.17	71.4 ± 0.26

Contributions

- Justin Tran: 4–11, 13–23
- Ethan Young: 1–3, 12, 24–38

References I

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